
ACTIVATIONS ARE NOT DEFAULTS: DATASET- AND REGIME-DEPENDENT NONLINEARITY SELECTION FOR LONG-HORIZON FORECASTING

Anonymous authors

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ABSTRACT

Activation functions in time-series forecasting models are usually treated as implementation defaults, while most methodological work focuses on architectures, attention mechanisms, input windows, and losses. We challenge this convention by studying activation-only interventions under fixed forecasting protocols. In a completed Informer full-grid study covering 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration, non-GELU activations are best in 34 of 36 cells; bounded and tanh-family activations achieve the strongest average ranks. These results show that activation choice is a first-order design variable even when attention, width, depth, optimizer, and training schedule are held fixed. The same grid also rejects a universal-activation story: Solar3 remains GELU-best at both horizons, output tanh is highly beneficial in selected long-horizon ETT/ETTm cells but weak on average, and decoder-only tanh wins no cell. Transfer stress tests sharpen this boundary. In PatchTST, non-GELU activations are best in 5 of 12 tested cells, but GELU remains best in 7 cells, and output tanh loses all 54 paired comparisons against matched linear heads. In a one-seed iTransformer pilot, GELU remains best in 5 of 6 ETTh2/ETTm2 cells, with only a narrow `tanh_sin001` win on ETTm2/96. We therefore frame activation selection as a conditional and architecture-dependent problem over activation family, placement, dataset, horizon, and input-window regime. The remaining submission gap is prospective factor-conditioned selection of activation regime and placement.

1 INTRODUCTION

Long-horizon forecasting papers often frame progress as architectural: more efficient attention, patching, decomposition, channel handling, or improved temporal embeddings. In contrast, the nonlinearity inside feed-forward blocks is commonly inherited from the reference implementation. This is a strong implicit assumption. A forecasting model’s activation function controls saturation, sign preservation, gradient scale, and sensitivity to large input excursions. These properties are directly relevant to time-series data, where volatility, jumps, trends, reversals, and heavy tails vary across datasets and horizons.

This paper studies a deliberately narrow question: if the forecasting architecture, optimizer, data split, horizon-specific window, and training schedule are fixed, can changing only the activation function and its placement materially improve performance? Our completed answer for Informer is yes. Across ETTh1, ETTh2, ETTm1, ETTm2, and eight Solar sites, we evaluate 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration. Non-GELU activations are best in 34 of 36 cells. The best average ranks belong to tanh-family and bounded activations: `tanh_sin001_all`, `tanh_all`, and `softsign_all`. These gains are not confined to one dataset: all 20 ETT/ETTm cells and 14 of 16 Solar cells prefer a non-GELU activation.

At the same time, our evidence argues against a universal activation. The best configuration varies across dataset and horizon. Output tanh wins several long-horizon ETT/ETTm cells but is weak globally. Encoder-side tanh is strongest in selected Solar96 cells. Decoder-only tanh never wins.

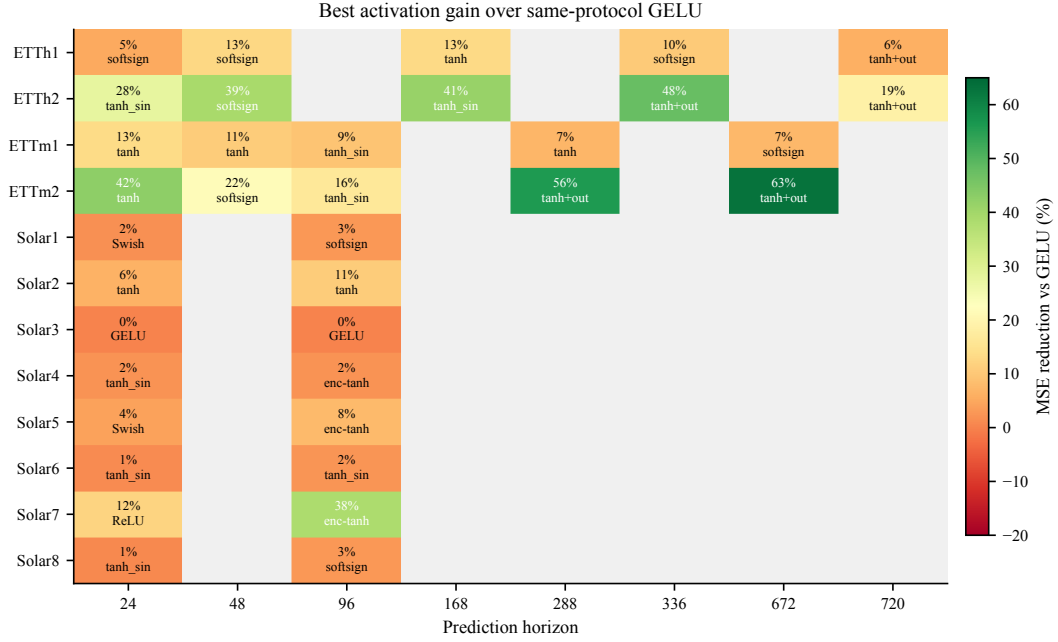


Figure 1: Best activation gain over the same-protocol GELU row in each completed dataset-horizon cell. The full grid covers 36 dataset-horizon cells, nine activation configurations, and three seeds. Most cells prefer a non-GELU activation, but Solar3 remains GELU-best at both horizons.

Solar3 remains GELU-best at both tested horizons. The scientific story is therefore not that we discovered a single replacement for GELU, but that activation choice is conditional.

The natural next question is whether this conditionality can be predicted from the data itself. We explore this using factor-bin diagnostics inspired by financial regime analysis. Input windows are summarized by volatility, realized volatility, jump intensity, trend consistency, max absolute change, tail ratio, reversal rate, mean reversion strength, and related factors. The goal is to determine whether these factors can predict both the activation regime and its placement, enabling low-cost activation selection without modifying the forecasting architecture.

The present evidence is not yet sufficient to claim a deployed zero-shot selector. Summary-level factor bins reveal regime dependence across the full matrix, but the stricter leave-one-seed-out oracle is positive only in Solar1/96. Therefore, this draft treats factor bins as a mechanism and selection-design tool until a prospective router experiment is completed. The PatchTST and iTransformer probes are now filled as architecture-transfer stress tests: they show activation sensitivity in selected cells, but they do not support broad portability of the Informer ranking.

Contributions. This draft organizes the paper around six contributions. First, we isolate activation family and placement in Informer while holding attention, width, depth, optimizer, data splits, and training protocol fixed. Second, we provide a complete three-seed Informer matrix of 972 seed-runs across 36 dataset-horizon cells and nine activation configurations. Third, we show that bounded and tanh-family activations are the strongest static candidates, while the complete grid rejects a universal winner. Fourth, we disentangle placement effects by separating full-hidden replacement, encoder-only tanh, decoder-only tanh, and hidden-plus-output tanh. Fifth, we introduce factor-bin diagnostics and specify the prospective experiment required to test whether input-window factors can select activation regime and placement. Sixth, we run PatchTST and iTransformer transfer stress tests showing that activation sensitivity can persist outside Informer, but the best family and placement are architecture-dependent.

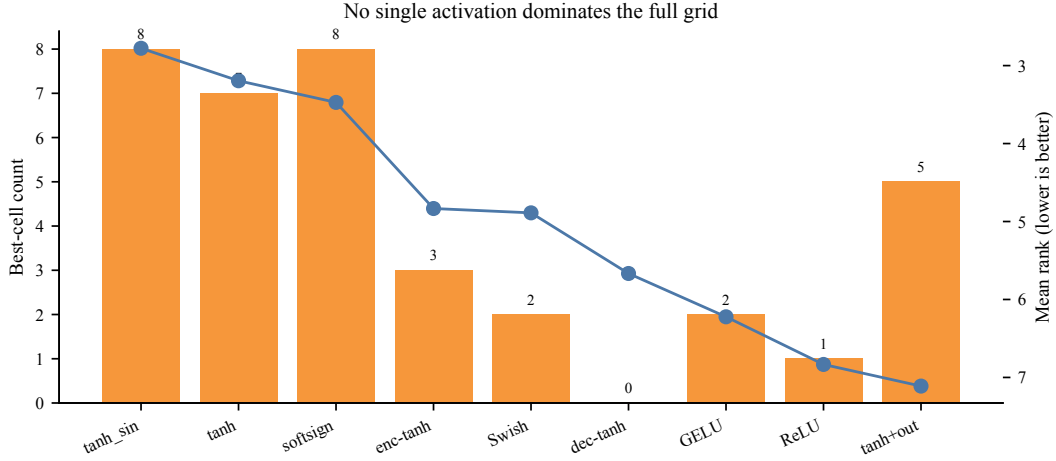


Figure 2: Static activation ranking across the full Phase-4 grid. Bounded and tanh-family activations rank best on average, but the best-cell counts are spread across several configurations, arguing against a universal replacement activation.

2 RELATED WORK

Transformer forecasting architectures. Informer (Zhou et al., 2021) adapts transformer sequence modeling (Vaswani et al., 2017) to long-sequence forecasting through ProbSparse attention and an encoder-decoder design. PatchTST (Nie et al., 2023) represents a different architectural family, using patch-level temporal tokens and channel-independent modeling. iTransformer (Liu et al., 2023) inverts the tokenization view by treating variates as tokens. Our completed evidence uses Informer as the controlled full-grid substrate and uses PatchTST and iTransformer as architecture-transfer stress tests.

Activation functions. ReLU (Nair & Hinton, 2010) and GELU (Hendrycks & Gimpel, 2016) are common defaults in deep networks. Bounded activations such as tanh and softsign are older, but they impose qualitatively different behavior: they preserve sign while constraining magnitude. Our results show that these properties can matter in forecasting even when no architectural component changes.

Oscillatory and Fourier-style nonlinearities. Periodic activations and Fourier features have been studied for representing high-frequency functions (Sitzmann et al., 2020; Tancik et al., 2020). Our use of oscillatory activations is more conservative: we test a small tanh-centered sinusoidal perturbation inside a forecasting transformer. The current result is that this perturbation is a strong static candidate, not that oscillation should be selected in every volatile regime.

Regime-conditioned analysis. Forecasting errors are often regime-dependent. We therefore bind test windows to input factors such as volatility, jumps, trend consistency, and reversals. This paper distinguishes two levels of evidence: post hoc factor-bin diagnostics, which reveal where activations differ, and prospective factor-conditioned routing, which must choose activation and placement before seeing test errors.

3 HYPOTHESES

H1: Activation functions are first-order design variables. Changing only activation family or placement, while holding architecture and protocol fixed, can materially change forecasting performance. This hypothesis is supported in Informer by 972 completed seed-runs and 34/36 non-GELU best cells.

H2: Bounded and tanh-family activations are strong static candidates. Bounded/sign-preserving nonlinearities and small tanh-centered oscillatory perturbations can outperform GELU-style defaults in many forecasting regimes. This hypothesis is supported as a static recommendation, but not as a universal winner.

H3: Oscillatory perturbations help in some regimes, but are not universally tied to volatility. `tanh_sin001_all` has the best mean rank in the Informer grid, but the current factor-bin evidence does not justify a direct rule such as “high volatility implies oscillatory activation.” A prospective factor-router experiment is required.

H4: Placement matters and can reverse the effect of the same activation. Full-hidden replacement, encoder-only tanh, decoder-only tanh, and output tanh have different empirical profiles. This hypothesis is supported: full-hidden placement wins 28 cells, hidden-plus-output tanh wins five, encoder-only tanh wins three, and decoder-only tanh wins none.

H5: Input-window factors can predict activation regime and placement. This is the central open hypothesis. Current factor-bin diagnostics show regime dependence, but they do not yet prove predictive selection. The final claim requires a rule that uses only input-window factors and is evaluated on held-out seeds, sites, horizons, or future windows.

4 METHOD

4.1 ACTIVATION-ONLY INTERVENTIONS

Let $X_{t-L+1:t} \in \mathbb{R}^{L \times C}$ denote an input window and let the model predict $Y_{t+1:t+H}$. We define an activation-only intervention as changing the nonlinearity used inside the forecasting model while keeping the architecture, attention, width, depth, optimizer, training schedule, data split, and horizon-specific windows fixed. In Informer, the intervention is applied to encoder and decoder feed-forward blocks, with an optional output activation used only for explicit placement tests.

Table 1: Activation configurations used in the completed Informer grid.

Config	Encoder FFN	Decoder FFN	Output	Purpose
<code>gelu_all</code>	GELU	GELU	Linear	Same-protocol baseline
<code>relu_all</code>	ReLU	ReLU	Linear	Monotone non-bounded control
<code>swish_all</code>	Swish	Swish	Linear	Smooth non-bounded control
<code>tanh_all</code>	tanh	tanh	Linear	Bounded sign-preserving candidate
<code>softsign_all</code>	softsign	softsign	Linear	Bounded sign-preserving candidate
<code>tanh_sin001_all</code>	$\tanh + 0.01 \sin$	$\tanh + 0.01 \sin$	Linear	Small oscillatory perturbation
<code>enc_tanh_dec_gelu</code>	tanh	GELU	Linear	Encoder-placement test
<code>enc_gelu_dec_tanh</code>	GELU	tanh	Linear	Decoder-placement test
<code>tanh_all_outtanh</code>	tanh	tanh	tanh	Hidden plus output-bounding test

4.2 FACTOR-BIN DIAGNOSTICS

For each input window, we compute regime factors: volatility, realized volatility, volatility shock, max absolute change, jump intensity, tail ratio, trend consistency, reversal rate, mean reversion strength, range, skewness, kurtosis, and downside/upside volatility. We use these factors in two ways. Summary-level factor bins group dataset-horizon cells and cover the full completed matrix; they are diagnostic but not deployable. Sample-level oracles use saved prediction arrays to estimate the benefit of choosing different activations in different factor bins. Same-split oracles are upper bounds. Leave-one-seed-out oracles provide weak generalization evidence.

4.3 PROSPECTIVE FACTOR ROUTER

The missing mechanism experiment is a prospective router. The router can use only input-window factors and must select both activation regime and placement before evaluation. Its action space should remain small enough to validate: `GELU`, `tanh_sin001_all`, `tanh_all`,

`softsign_all`, `enc_tanh_dec_gelu`, and `tanh_all_outtanh`. It must be compared against GELU, the best static activation, and the same-split oracle upper bound.

Table 2: Candidate prospective routers. Results are intentionally left pending in this hypothesis draft.

Router	Description	Claim strength if successful
Rule-based threshold	Hand-designed thresholds on volatility, jumps, and trend factors	Weak but interpretable
Cell-trained decision tree	Learn factor-to-activation rules on training cells	Moderate
Leave-dataset-out router	Train on all but one dataset/site, test on held-out site	Strong
Leave-horizon-out router	Train on some horizons, test on unseen horizon	Strong
Uncertainty-gated router	Choose best static activation unless confidence is high	Most practical

5 COMPLETED INFORMER EVIDENCE

5.1 FULL-GRID COVERAGE

The completed Informer Phase-4 matrix contains 972 eligible seed-runs: 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration. All 324 dataset-horizon-configuration cells have complete three-seed coverage under batch size 32 and six epochs.

5.2 ACTIVATION-ONLY CHANGES BEAT GELU IN MOST CELLS

Figure 1 summarizes the main result. Non-GELU activations are best in 34 of 36 cells. The only GELU-best cases are Solar3 at horizons 24 and 96. ETT and ETTm cells show the largest gains: all 20 ETT/ETTm cells prefer non-GELU activations, with mean best-cell MSE reductions of 22.1% on ETTh and 24.7% on ETTm. Solar gains are smaller and more site-specific: 14 of 16 Solar cells prefer non-GELU activations.

The static ranking shows that the strongest average configurations are `tanh_sin001_all`, `tanh_all`, and `softsign_all`. This supports bounded/tanh-family activations as strong static candidates, but the best-cell wins are distributed across multiple configurations. This is the empirical reason to frame the result as conditional selection rather than universal replacement.

5.3 PLACEMENT IS CONDITIONAL

Placement effects are substantial. Full-hidden replacement is the most stable default, winning 28 of 36 cells. Hidden-plus-output tanh wins five long-horizon ETT/ETTm cells: ETTh1/720, ETTh2/336, ETTh2/720, ETTm2/288, and ETTm2/672. Encoder-only tanh wins Solar4/96, Solar5/96, and Solar7/96. Decoder-only tanh wins no cell. These results rule out the interpretation that all gains come from output clipping or any single insertion point.

5.4 FACTOR-BIN DIAGNOSTICS REVEAL DEPENDENCE BUT NOT ROUTING

Summary-level factor bins cover all 36 dataset-horizon cells and produce 783 factor-bin rows. They reveal many factor-conditioned contrasts, which indicates that activation differences vary with regime summaries. The stricter sample-level leave-one-seed-out oracle is much weaker: among 84 oracle rows, only seven are positive, all on Solar1/96.

Thus, factor bins are useful for mechanism diagnosis and experiment design, but the current evidence does not support a deployed router. The router must be validated prospectively.

Table 3: Phase-4 full-grid coverage. A main-table cell is eligible only when all three seeds are present under batch size 32 and six training epochs.

Group	dataset-horizon cells	config cells	seed-runs	3-seed config cells
ETTh	10	90	270/270	90/90
ETTm	10	90	270/270	90/90
Solar	16	144	432/432	144/144
All	36	324	972/972	324/324

Table 4: Static activation ranking across the 36 completed dataset-horizon cells. Rank is computed within each cell over the nine Phase-4 configurations.

Config	mean rank	best cells	top-3 cells	mean gain	positive cells
tanh_sin	2.78	8	27	+11.7%	83.3%
tanh	3.19	7	25	+11.9%	80.6%
softsign	3.47	8	22	+11.2%	80.6%
enc-tanh	4.83	3	11	+5.1%	66.7%
Swish	4.89	2	5	+3.6%	77.8%
dec-tanh	5.67	0	3	+1.9%	66.7%
GELU	6.22	2	4	+0.0%	0.0%
ReLU	6.83	1	4	-2.3%	36.1%
tanh+out	7.11	5	7	-36.3%	25.0%

Table 5: Top positive leave-one-seed-out factor-oracle diagnostics among array-ready Phase-4 cells. These are diagnostic upper bounds for regime sensitivity, not deployed activation-routing results.

Dataset	H	factor	LOSO oracle MSE	best static MSE	gain
Solar1	96	max_abs_change	0.174	0.178	+2.0%
Solar1	96	volatility	0.175	0.178	+1.4%
Solar1	96	volatility_shock	0.175	0.178	+1.3%
Solar1	96	trend_consistency	0.177	0.178	+0.4%
Solar1	96	max_drawdown	0.177	0.178	+0.2%
Solar1	96	max_drawup	0.177	0.178	+0.2%
Solar1	96	range	0.177	0.178	+0.2%

6 COMPLETED EXPERIMENT A: PATCHTST TRANSFER

6.1 GOAL

This experiment tests whether activation sensitivity transfers from Informer to PatchTST. The result is partial: activation choice still matters in selected cells, but the Informer static ranking and output-tanh placement do not directly transfer.

6.2 TRANSFER RESULTS

Interpretation. Non-GELU activations win 5 of 12 PatchTST cells, concentrated in ETTh2/720 and Solar2/Solar5. GELU remains best in 7 cells, including all tested ETTm2 horizons and both Solar3 horizons. Output tanh loses all 54 paired seed-run comparisons against matching linear-head configurations. The paper should therefore claim architecture-sensitive transfer: activation choice remains relevant outside Informer, but neither the Informer ranking nor output-bounding placement is portable as a default rule.

Table 6: PatchTST activation-transfer stress test. The Informer activation effect transfers only partially: non-GELU activations win selected ETTh2/Solar cells, while GELU remains best in 7 of 12 tested cells.

Model	Dataset	Horizon	GELU MSE	Best config	Gain
PatchTST	ETTh2	168	0.325880	gelu_ffn_linear	0.00%
PatchTST	ETTh2	336	0.328450	gelu_ffn_linear	0.00%
PatchTST	ETTh2	720	0.380881	softsign_ffn_linear	0.67%
PatchTST	ETTm2	96	0.165589	gelu_ffn_linear	0.00%
PatchTST	ETTm2	288	0.262739	gelu_ffn_linear	0.00%
PatchTST	ETTm2	672	0.355232	gelu_ffn_linear	0.00%
PatchTST	Solar2	24	0.065450	tanh_sin001_ffn_linear	0.42%
PatchTST	Solar2	96	0.099666	softsign_ffn_linear	1.12%
PatchTST	Solar3	24	0.279035	gelu_ffn_linear	0.00%
PatchTST	Solar3	96	0.446409	gelu_ffn_linear	0.00%
PatchTST	Solar5	24	0.114792	tanh_sin001_ffn_linear	1.33%
PatchTST	Solar5	96	0.176366	softsign_ffn_linear	1.01%

Table 7: PatchTST output-tanh placement test. Output tanh loses every paired seed-run comparison against the matching linear-head configuration, indicating that the Informer output-bounding signal does not transfer to PatchTST.

Base activation	Paired seed-runs	Win rate	Mean gain vs. linear
tanh	18	0.0%	-5.28%
tanh_sin001	18	0.0%	-5.31%
gelu	18	0.0%	-5.44%

7 COMPLETED EXPERIMENT B: iTRANSFORMER TRANSFER

7.1 GOAL

This experiment asks whether the same FFN activation families transfer to iTransformer. It is intentionally smaller than the PatchTST study: one seed, two ETT datasets, three horizons, and no output-placement variants. It should therefore be read as a fast architecture-transfer probe rather than a final multi-seed claim.

7.2 TRANSFER RESULTS

Interpretation. iTransformer is a stricter negative/weak transfer test. GELU remains best in 5 of 6 ETTh2/ETTm2 cells. The only non-GELU win is ETTm2/96, where `tanh_sin001_ffn_linear` improves MSE over GELU by 0.19%. This does not support direct migration of the Informer tanh-family ranking to iTransformer. Instead, together with the PatchTST result, it supports the narrower claim that activation effects are architecture-dependent and should be selected conditionally.

Table 8: iTransformer activation-transfer stress test. In this one-seed pilot, GELU remains best in 5 of 6 ETTh2/ETTM2 cells; the only non-GELU win is a narrow `tanh_sin001` gain on ETTm2/96.

Model	Dataset	Horizon	GELU MSE	Best config	Gain
iTransformer	ETTh2	96	0.301598	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTh2	336	0.422894	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTh2	720	0.428986	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTM2	96	0.184226	<code>tanh_sin001_ffn_linear</code>	0.19%
iTransformer	ETTM2	336	0.313418	<code>gelu_ffn_linear</code>	0.00%
iTransformer	ETTM2	720	0.414092	<code>gelu_ffn_linear</code>	0.00%

8 MISSING EXPERIMENT C: FACTOR-CONDITIONED SELECTION

8.1 GOAL

The second missing experiment tests whether input-window factors can prospectively predict activation regime and placement. This experiment determines whether the paper can claim a selection mechanism rather than only a diagnostic map.

8.2 VALIDATION MATRIX

Table 9: Prospective factor-router validation matrix.

Validation split	Train source	Test target	Required output
Leave-one-seed-out	Two seeds in same cell	Held-out seed	Router MSE vs best static
Leave-one-Solar-site-out	Solar sites except one	Held-out Solar site	Generalization across sites
Leave-one-horizon-out	Some horizons	Unseen horizon	Horizon transfer
Future-window split	Earlier test windows	Later test windows	Time-forward robustness
Leave-family-out	ETT train, Solar test or inverse	Held-out family	Cross-family robustness

Table 10: Planned prospective factor-router result table.

Split	Scope	GELU MSE	Best static MSE	Oracle MSE	Router MSE	Gain vs static
Leave-one-seed-out	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Leave-one-Solar-site-out	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Leave-one-horizon-out	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Future-window split	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Leave-family-out	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

Table 11: Planned mechanism table for factor-to-placement rules.

Factor condition	Selected activation	Selected placement	Held-out evidence	Use in final paper?
High volatility + high max absolute change	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
High trend consistency	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
High jump intensity	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
High reversal rate	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Low volatility + stable trend	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

Success criteria. The router must beat GELU. It should beat or match the best static activation on at least one held-out split. It should not collapse on counterexample cells such as Solar3. Its rules should be interpretable, because the claim is not merely that an oracle exists but that factors can guide activation and placement selection before evaluation.

9 DISCUSSION

The completed Informer grid already supports a meaningful paper: activations are not harmless defaults in long-horizon forecasting. The completed PatchTST and iTransformer stress tests narrow the generality claim: activation sensitivity is not purely an Informer artifact, but the activation ranking and placement policy are architecture-dependent. Prospective factor routing remains the key missing experiment that determines whether factor bins can move from diagnosis to prediction.

This draft therefore uses a claim ladder. The safe claim is that activation choice is dataset- and horizon-sensitive in Informer. The cross-architecture claim is weaker but still useful: activation sensitivity remains visible in selected PatchTST and iTransformer cells, while the best activation family and placement are architecture-dependent. The strongest claim, only after router validation, is that activation regime and placement can be selected from data characteristics to improve forecasting without modifying the model architecture.

10 CONCLUSION

Our current evidence shows that activation-only interventions can substantially change Informer forecasting performance under fixed protocols. Non-GELU activations win 34 of 36 dataset-horizon cells, and bounded/tanh-family activations are the best static candidates on average. Yet the same evidence rejects a universal activation: placement, dataset, horizon, regime, and architecture all matter. PatchTST confirms that transfer is partial and output tanh fails consistently; iTransformer further shows that GELU can remain the strongest default in a different transformer variant. The final submission should therefore be built around conditional activation selection. The remaining work is sharply defined: test whether factor bins can prospectively select activation regime and placement.

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Table 12: Best activation per dataset-horizon in the completed Phase-4 grid. Improvements are relative MSE reductions against the same-protocol GELU row with matched seeds; GELU-best cells have no relative-improvement entry.

Dataset	H	best config	MSE	gain vs GELU	seed win rate
ETTh1	24	softsign	0.532 ± 0.026	+5.2%	66.7%
ETTh1	48	softsign	0.646 ± 0.073	+12.6%	100.0%
ETTh1	168	tanh	0.912 ± 0.131	+12.9%	66.7%
ETTh1	336	softsign	1.219 ± 0.042	+10.4%	100.0%
ETTh1	720	tanh+out	1.328 ± 0.030	+6.0%	100.0%
ETTh2	24	tanh_sin	0.516 ± 0.023	+27.8%	100.0%
ETTh2	48	softsign	1.606 ± 0.386	+38.8%	100.0%
ETTh2	168	tanh_sin	2.963 ± 0.509	+41.3%	100.0%
ETTh2	336	tanh+out	2.054 ± 0.219	+47.7%	100.0%
ETTh2	720	tanh+out	3.298 ± 0.186	+18.6%	100.0%
ETTh1	24	tanh	0.314 ± 0.015	+13.4%	100.0%
ETTh1	48	tanh	0.453 ± 0.023	+10.8%	100.0%
ETTh1	96	tanh_sin	0.559 ± 0.050	+9.3%	100.0%
ETTh1	288	tanh	0.802 ± 0.044	+7.2%	66.7%
ETTh1	672	softsign	0.922 ± 0.023	+7.3%	66.7%
ETTh2	24	tanh	0.685 ± 0.191	+42.5%	100.0%
ETTh2	48	softsign	0.309 ± 0.014	+21.6%	100.0%
ETTh2	96	tanh_sin	1.335 ± 0.214	+16.4%	100.0%
ETTh2	288	tanh+out	2.355 ± 0.359	+55.7%	100.0%
ETTh2	672	tanh+out	2.851 ± 0.162	+62.7%	100.0%
Solar1	24	Swish	0.150 ± 0.000	+1.7%	66.7%
Solar1	96	softsign	0.224 ± 0.001	+2.6%	100.0%
Solar2	24	tanh	0.066 ± 0.003	+6.4%	100.0%
Solar2	96	tanh	0.108 ± 0.005	+10.8%	100.0%
Solar3	24	GELU	0.198 ± 0.009	—	—
Solar3	96	GELU	0.349 ± 0.007	—	—
Solar4	24	tanh_sin	0.220 ± 0.003	+1.9%	100.0%
Solar4	96	enc-tanh	0.290 ± 0.005	+1.9%	66.7%
Solar5	24	Swish	0.109 ± 0.002	+4.2%	100.0%
Solar5	96	enc-tanh	0.180 ± 0.003	+7.6%	100.0%
Solar6	24	tanh_sin	0.265 ± 0.002	+1.1%	100.0%
Solar6	96	tanh_sin	0.358 ± 0.009	+2.3%	100.0%
Solar7	24	ReLU	3.281 ± 1.035	+12.0%	66.7%
Solar7	96	enc-tanh	4.970 ± 1.353	+38.2%	100.0%
Solar8	24	tanh_sin	0.216 ± 0.003	+0.8%	100.0%
Solar8	96	softsign	0.306 ± 0.005	+2.6%	100.0%

B PENDING EXPERIMENT SUMMARY

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Table 13: Final missing experiments before the strongest submission claim.

Experiment	Purpose	Minimum size	Claim enabled if successful
PatchTST transfer	Test architecture transfer	144 + 54 runs	Completed: partial, architecture-dependent transfer
iTransformer transfer	Test architecture transfer	24 runs	Completed: weak/negative direct transfer
Prospective factor router	Test factor-to-activation prediction	Split-dependent	Data factors can select activation regime and placement